

Multi-Member Districts via Recursive Bisection: STV and OLPR Variants

Giovanni Della-Libera

May 7, 2026

Abstract

The United States House of Representatives is elected exclusively through single-member plurality districts, a design that enables partisan gerrymandering by allowing mapmakers to pack and crack opposition voters across district boundaries. Multi-member districts (MMDs) eliminate this vulnerability by replacing geographic seat allocation with within-district proportional representation: even a biased boundary cannot deny a party that wins 40% of a district’s votes its proportional share of seats. We demonstrate that the bisect recursive bisection algorithm can generate MMDs by adjusting the seat-count parameter k —for example, dividing California’s 52 seats into 13 four-member districts ($k = 13$) or New Jersey’s 12 seats into 6 two-member districts ($k = 6$). We implement two leading MMD seat-allocation rules—Single Transferable Vote (STV) and Open-List Proportional Representation (OLPR)—and evaluate them on three state case studies (California, New Jersey, Illinois) using 2020 presidential election data. MMDs systematically improve minority representation relative to single-member districts: California’s 13 four-member districts produce an estimated 21–23 minority-opportunity seats, compared to 17 majority-minority districts in the enacted single-member plan.[†] Compactness (Polsby-Popper) declines modestly—approximately 8% relative to single-member tract-level redistricting—as larger districts span more varied terrain.[†] These results establish MMD generation as a practical extension of the bisect framework and provide a computational baseline for electoral reform evaluation.

Keywords: multi-member districts, proportional representation, single transferable vote, open-list proportional representation, recursive bisection, electoral reform, algorithmic redistricting

1 Introduction

Congressional redistricting in the United States operates under a fundamental structural constraint: 435 single-member districts elected by winner-take-all plurality voting. This combination creates systematic incentives for partisan manipulation. A party that controls the redistricting process can engineer district boundaries to waste opposition votes—packing them into a few overwhelmingly safe districts, or cracking them across many districts where they form ineffective minorities [15]. The result is a persistent gap between a party’s

statewide vote share and its seat share, a gap that is structurally guaranteed to favour whichever party drew the map.

Multi-member districts (MMDs) break this dynamic at the source. When a district elects four representatives and seats are allocated proportionally within the district, a party that wins 40% of the district’s votes receives approximately 40% of its seats. No boundary can be drawn to deny a large minority party its seats once the allocation rule is changed from winner-take-all to proportional. McGann et al. [12] argue that MMDs are structurally gerrymandering-proof: the proportional allocation mechanism makes packing and cracking strategies geometrically ineffective.

The United States has historical experience with MMDs. Illinois elected state legislators from three-member districts using Cumulative Voting from 1870 to 1980 [4]. Several U.S. cities (Cambridge, Massachusetts; Portland, Maine) currently use STV for municipal elections. The Federal Apportionment Act of 1842 was the first statute to mandate single-member districts for U.S. House elections; the Constitution itself does not require single-member districts.

1.1 This Paper

We ask a concrete algorithmic question: can the bisect recursive bisection framework generate multi-member congressional districts, and if so, what are the compactness and representation consequences? We answer affirmatively. Bisect generates MMDs by adjusting the district count k :

- For California (52 seats), $k = 13$ districts of 4 seats each, or $k = 26$ districts of 2 seats each.
- For New Jersey (12 seats), $k = 6$ districts of 2 seats each, or $k = 4$ districts of 3 seats each.
- For Illinois (17 seats), $k = 5$ districts of 3 seats each (with 2 districts of 4 seats for exact seat coverage, or using a mixed-magnitude design).

We implement two seat-allocation rules within each MMD: Single Transferable Vote (STV, also known as Proportional Ranked Choice) and Open-List Proportional Representation (OLPR, the system used in Sweden, Finland, and Brazil). These two systems represent the major MMD design choices available: STV is voter-centric (voters rank individual candidates), while OLPR is party-centric (voters choose a party list with an optional candidate preference).

1.2 Why Algorithmic MMD Generation Matters

Computational redistricting tools have focused almost entirely on single-member districts. The algorithmic MMD question has been left to political scientists, who typically draw illustrative examples by hand [14]. Our contribution is to place MMD generation on the same principled, reproducible algorithmic footing as single-member redistricting. This enables:

1. *Systematic comparison*: MMD and single-member plans generated by the same algorithm from the same data, making the district-magnitude effect cleanly separable from boundary-drawing effects.
2. *Legal baselines*: Courts and commissions evaluating MMD reform proposals need computational plans to compare against; we provide the methodology.
3. *VRA analysis*: The Voting Rights Act imposes specific requirements on minority representation; MMDs interact with these requirements in complex ways that require systematic analysis.

1.3 Paper Structure

Section 2 reviews STV and OLPR mechanics and the MMD compactness literature. Section 3 describes the bisect k -adjustment for MMDs and the per-district balance-tolerance formula. Section 4 presents results for the three state case studies. Section 5 addresses the constitutional and statutory legal landscape. Section 6 concludes.

2 Background

2.1 Single Transferable Vote (STV)

STV is a ranked-choice proportional system in which voters rank individual candidates from all parties in preference order. Seats are filled through an iterative elimination-and-transfer process [8]:

1. Compute the Droop quota: $Q = \lfloor P/(S+1) \rfloor + 1$, where P is total valid ballots and S is the number of seats.
2. Candidates who reach quota are elected; their surplus votes (above quota) are redistributed to next preferences at a transfer value $TV = \text{surplus}/\text{total votes for candidate}$.
3. If no candidate reaches quota, eliminate the candidate with the fewest votes and redistribute their votes at full value.
4. Repeat until all seats are filled.

STV produces proportional outcomes when parties are cohesive in their voter rankings. Its main advantage over closed-list PR is that voters can express preferences within parties, not just between parties. Its main disadvantage is ballot complexity: in a 4-member district with 5 parties each running 2 candidates, a ballot may rank up to 10 candidates.

STV is used in Ireland (Dail Eireann, 3–5 member constituencies), Malta (5 members), Scotland (local government, 3–4 members), Australia (Senate, state-by-state, 6–12 members), and several U.S. and Canadian cities. The Irish system, with 3–5 member constituencies, is the closest international model to a potential U.S. House reform.

2.2 Open-List Proportional Representation (OLPR)

OLPR is a party-list system in which voters choose a party and may additionally cast a candidate preference vote [11]. Seats are allocated to parties using a divisor method (D’Hondt or Sainte-Lague), and within each party, seats go to candidates with the most preference votes. If a voter does not express a candidate preference, their vote counts toward the party’s total but not toward any specific candidate.

OLPR is used in Sweden, Finland, the Netherlands, and Brazil (Chamber of Deputies). The defining feature is that voters can influence which individuals from their preferred party are elected, while the party-list structure ensures that party seat shares are proportional to vote shares.

For algorithmic implementation, OLPR is simpler than STV: the seat-allocation step is a D’Hondt divisor calculation, well-established in the literature [3]. The candidate-preference ranking within parties is a separate computation that we implement as a post-processing step.

2.3 District Magnitude and Compactness

District magnitude—the number of seats per district—is the central design variable in MMD systems. Higher magnitude improves proportionality (lower Gallagher index) but tends to produce less compact districts, because larger population districts must span more territory.

The relationship between magnitude and compactness has been studied theoretically [16] but rarely empirically for the United States. Altman and McDonald [2] noted that algorithmic redistricting produces more compact districts as k increases (more, smaller districts), which is the opposite direction from MMD design (fewer, larger districts).

For a fixed total population P and k districts of magnitude M (so $kM = S$ total seats), the mean district population is P/k . A district with twice the population of a single-member district is not necessarily twice as spatially spread, because population is concentrated—but in low-density states, larger districts do span more land, increasing perimeter and reducing Polsby-Popper.

2.4 MMDs and Minority Representation

Under single-member districts, the primary mechanism for minority representation is the majority-minority district (MMD): a district where a minority group comprises $> 50\%$ of the voting-age population, enabling the election of minority-preferred candidates [17]. The Voting Rights Act (Section 2) can require such districts when specific legal conditions are met [18].

Under multi-member proportional districts, minority representation can occur without geographic concentration. A district where a minority group is 35% of the population electing 4 representatives would typically elect 1–2 minority-preferred candidates under proportional allocation. This “representation without packing” was argued by Guinier [10] as a superior approach to the geographic concentration required by majority-minority single-member districts.

The tradeoff is that MMD minority representation is contingent on minority voting cohesion and the availability of minority candidates across party lists. Geographic concentration in a single-member district guarantees minority representation; proportional allocation in an MMD converts minority vote share into seat share only when the minority votes cohesively for candidates of their choosing.

3 Algorithm

3.1 Bisect k -Adjustment for MMDs

The bisect algorithm generates a k -partition of a state’s census tracts into equal-population districts. For single-member redistricting, k equals the number of congressional seats apportioned to the state. For MMDs, k is reduced to the number of districts, which is S/M for a uniform-magnitude design (where S is total seats and M is seats per district):

$$k_{\text{MMD}} = \left\lfloor \frac{S}{M} \right\rfloor \quad (1)$$

For non-divisible cases (e.g., Illinois with $S = 17$, $M = 3$: $k = 5$ with one district taking 4 seats and four districts taking 3), we implement a mixed-magnitude design where the algorithm generates 5 equal-population partitions and the seat count per district is assigned post-hoc based on population ranking.

The bisect CLI exposes this through the `--seats-per-district` flag:

```
bisect state --state CA --year 2020 --version E1_mmd4 \
  --districts 13 --seats-per-district 4 \
  --balance-tolerance 5.0 --seed 42
```

3.2 Per-District Balance Tolerance

A critical algorithmic adjustment for MMDs is the balance tolerance. Single-member redistricting targets $\pm 0.5\%$ population deviation from the ideal district population (P/k where P is state population). This strict tolerance reflects the “one person, one vote” constitutional standard for congressional districts [19].

For MMDs, the relevant equality standard is *per seat*, not per district: equal representation means each seat represents the same number of people, regardless of how seats are grouped into districts. For a k -district design with M seats per district:

$$\text{ideal per seat} = \frac{P}{k \cdot M} = \frac{P}{S} \quad (2)$$

$$\text{ideal per district} = M \cdot \frac{P}{S} \quad (3)$$

The tolerance per district is therefore M times the per-seat tolerance. If we target $\pm 0.5\%$ per seat, the per-district tolerance is $\pm 0.5\% \cdot M$. For 4-member districts this is $\pm 2.0\%$; for 2-member districts, $\pm 1.0\%$.

However, this relaxed tolerance is appropriate for the population balance constraint only if the *per-seat* standard is applied when evaluating constitutional compliance. We implement both: the algorithm uses the per-district tolerance for feasibility, and we report per-seat deviations in our results.

3.3 METIS Ufactor Scaling

The METIS partitioner uses an “imbalance factor” (ufactor) to control population balance at each bisection node. As described in prior portfolio work [5], the ufactor must be scaled by district count to prevent compounding balance errors at each bisection level.

For MMD generation, the ufactor formula is:

$$\text{ufactor}_{\text{node}} = 1 + \frac{T \cdot M}{k_{\text{node}}} \quad (4)$$

where T is the per-seat tolerance (0.005 for $\pm 0.5\%$), M is seats per district, and k_{node} is the number of districts in the current subtree. This guarantees cumulative balance error $\leq T \cdot M$ regardless of bisection depth.

3.4 STV Implementation

We implement the Gregory method of STV [8], which uses fractional surplus transfer:

1. Compute Droop quota $Q = \lfloor P/(S + 1) \rfloor + 1$.
2. Award seats to all candidates with votes $\geq Q$. Transfer surplus at value $TV = (v_c - Q)/v_c$ applied to all continuing ballots for that candidate.
3. If no candidate reaches Q , eliminate the candidate with fewest first-preference votes. Transfer their votes at full value.
4. Repeat until all S seats are filled or remaining candidates equal remaining vacancies.

For simulation purposes (where we do not have individual ranked ballots), we use vote-share input: parties receive their seat-proportion of the Droop quota from aggregate vote data. This approximation is standard in comparative electoral studies [7] and produces accurate seat counts when parties are cohesive.

3.5 OLPR Implementation

OLPR seat allocation uses the D’Hondt highest-averages method:

1. For each party p with vote total v_p , compute quotients $v_p/(s_p + 1)$ where s_p is the current seat count.
2. Award the next seat to the party with the highest quotient.
3. Repeat until all S seats are awarded.

The effective threshold—the minimum vote share needed to win at least one seat—is approximately $1/(S + 1)$ (Droop quota): 20% for 4-member districts, 25% for 3-member, 33% for 2-member.

3.6 Data and Configurations

We run three state case studies using 2020 census tract data and 2020 presidential election returns [13]:

California ($S = 52$). Two configurations: $k = 13$ districts of 4 seats (“CA-4”), and $k = 26$ districts of 2 seats (“CA-2”).

New Jersey ($S = 12$). Two configurations: $k = 4$ districts of 3 seats (“NJ-3”), and $k = 6$ districts of 2 seats (“NJ-2”).

Illinois ($S = 17$). One configuration: $k = 5$ districts with a mixed magnitude (three 3-seat and two 4-seat districts, “IL-mix”).

All runs use seed 42, geographic edge-weighting, and the `single` search mode. Results carry the single-run dagger ([†]) throughout.

4 Results

4.1 District Generation Summary

Table 1 summarises the five MMD configurations generated across three states. All configurations achieve population balance within the per-seat $\pm 0.5\%$ standard.[†]

Table 1: MMD configurations generated by bisect (2020 Census, seed 42).[†]

State	Config	Districts	Seats/District	Mean Pop	Per-Seat Dev
CA	CA-4	13	4	3.0M	$\pm 0.4\%$
CA	CA-2	26	2	1.5M	$\pm 0.4\%$
NJ	NJ-3	4	3	2.7M	$\pm 0.3\%$
NJ	NJ-2	6	2	1.8M	$\pm 0.3\%$
IL	IL-mix	5	3/4	2.5M	$\pm 0.5\%$

4.2 Compactness Results

Table 2 reports Polsby-Popper scores for each MMD configuration alongside the bisect single-member baseline for the same state.

Table 2: Polsby-Popper compactness: MMD vs. single-member baseline.[†]

State	Config	Mean PP	Vs. SM Baseline	PP ≥ 0.3
CA	Single-member ($k = 52$)	0.182	–	52%
CA	CA-4 ($k = 13$)	0.161	–11.5%	38%
CA	CA-2 ($k = 26$)	0.172	–5.5%	44%
NJ	Single-member ($k = 12$)	0.241	–	75%
NJ	NJ-3 ($k = 4$)	0.218	–9.5%	50%
NJ	NJ-2 ($k = 6$)	0.229	–5.0%	67%
IL	Single-member ($k = 17$)	0.198	–	47%
IL	IL-mix ($k = 5$)	0.175	–11.6%	20%

Compactness declines monotonically as district count decreases (magnitude increases). The 2-member configurations (CA-2, NJ-2) lose approximately 5–6% in mean PP; the 4-member configuration (CA-4) loses approximately 11–12%. This is consistent with the theoretical expectation: larger districts span more varied terrain, increasing perimeter relative to area.

The pattern is qualitatively the same across all states, but the magnitude of loss varies. California, with its complex Pacific coastline and Sierra Nevada boundary, experiences a larger absolute PP reduction per unit of district-count reduction than New Jersey, which has a more compact geography.

4.3 Minority Representation

Table 3 estimates minority-opportunity seats under each configuration using 2020 presidential election returns as a partisan proxy and census minority population share as a demographic proxy.

Table 3: Estimated minority-opportunity seats: MMD vs. single-member baseline.[†]

State	Config	Minority-Opp. Seats	vs. Single-Member
CA	Enacted (single-member)	17	–
CA	CA-4 (STV)	21	+4
CA	CA-4 (OLPR)	22	+5
CA	CA-2 (STV)	19	+2
NJ	Enacted (single-member)	3	–
NJ	NJ-3 (STV)	5	+2
NJ	NJ-2 (STV)	4	+1
IL	Enacted (single-member)	4	–
IL	IL-mix (OLPR)	6	+2

All MMD configurations produce more minority-opportunity seats than the enacted

single-member plans.[†]The improvement is largest for California’s 4-member districts: OLPR produces an estimated 22 minority- opportunity seats versus 17 in the enacted plan, a 29% improvement. The mechanism is “representation without packing”: several districts where minority populations are 25–40% of the total can elect minority-preferred candidates under proportional allocation without requiring the geographic concentration that creates majority-minority single-member districts.

4.4 STV vs. OLPR Comparison

For the California case study (most seats, clearest signal), OLPR consistently produces one more minority-opportunity seat than STV at equivalent district magnitude.[†]This reflects a structural difference: OLPR allows voters to give preference votes directly to minority candidates on the party list, concentrating their influence even when the overall minority share of district population is below 50%. STV requires minorities to rank minority-preferred candidates first *and* to have their transfers flow to minority candidates at subsequent preferences, which is probabilistically less efficient.

Both systems substantially outperform single-member districts for minority representation, confirming the theoretical prediction from Guinier [10].

4.5 Proportionality

Using 2020 presidential vote shares as a proxy for party vote shares, we compute the Gallagher index for each configuration:

Table 4: Gallagher proportionality index by configuration (lower = more proportional).[†]

State	Config	Gallagher Index
CA	Enacted (single-member)	9.2
CA	CA-4	1.8
CA	CA-2	3.1
NJ	Enacted (single-member)	7.4
NJ	NJ-3	2.3
NJ	NJ-2	4.1
IL	Enacted (single-member)	8.7
IL	IL-mix	2.9

All MMD configurations achieve Gallagher indices well below the single-member baselines, with 4-member districts (CA-4) reaching a Gallagher index of 1.8— comparable to Nordic proportional representation systems [9].[†]The 2-member configurations remain more proportional than single-member plurality but are less proportional than the larger-magnitude options, consistent with the effective-threshold theory: a 2-member district has a Droop-quota threshold of 33%, excluding parties with less than one-third of district votes.

5 Legal Landscape

5.1 Constitutional Status of MMDs

The U.S. Constitution does not require single-member districts for the House of Representatives. Article I, Section 2 requires only that representatives be “chosen every second Year by the People of the several States,” and Section 4 gives Congress power to regulate the “Time, Places and Manner” of elections. The Constitution is silent on district magnitude.

The statutory requirement for single-member districts derives from 2 U.S.C. § 2c, enacted in its current form in 1967. This statute reads: “In each State entitled in the Ninety-first Congress or in any subsequent Congress to more than one Representative under an apportionment made pursuant to the provisions of [the apportionment statutes], there shall be established by law a number of districts equal to the number of Representatives to which such State is so entitled, and Representatives shall be elected only from districts so established.” This language clearly requires single-member districts as a matter of current statutory law. However, Congress could repeal or amend Section 2c. Bills to do so have been introduced repeatedly [1].

5.2 Illinois Precedent (1870–1980)

The most relevant domestic precedent for MMD redistricting is Illinois’s state legislative system from 1870 to 1980. Under the Illinois Constitution of 1870, the state Senate was elected from single-member districts while the House was elected from three-member districts using Cumulative Voting [4]. Each voter cast three votes, allocable in whole or fractional amounts among candidates.

Cumulative Voting in Illinois produced significant minority party representation: Republicans routinely won one of three seats in safe Democratic Chicago districts, and Democrats won one seat in safe Republican downstate districts. The system was abolished by referendum in 1980 (the “Cutback Amendment”) under pressure from a taxpayer revolt seeking to reduce the size of the legislature rather than because of dissatisfaction with its representative outcomes [4].

The Illinois precedent is directly relevant because: (a) it demonstrates the constitutional validity of MMDs at the state legislative level; (b) it provides 110 years of operational experience with three-member districts in an American political context; and (c) the elimination of the system was motivated by cost-cutting, not representational failure.

5.3 Voting Rights Act Implications

Section 2 of the Voting Rights Act (VRA) prohibits voting practices that “result in a denial or abridgement of the right ... to vote on account of race or color.” Under *Thornburg v. Gingles* [18], a Section 2 violation requires showing that (1) the minority group is sufficiently large and geographically compact to constitute a majority in a single-member district, (2) the minority is politically cohesive, and (3) the white majority votes sufficiently as a bloc to defeat minority-preferred candidates.

The interaction between MMDs and the VRA is complex. On one hand, MMDs may satisfy VRA requirements without creating majority-minority single-member districts: proportional allocation in a 4-member district where minorities are 35% of the population will typically elect 1–2 minority-preferred candidates. On the other hand, *Thornburg’s* “sufficiently compact” prong has been interpreted to require geographic compactness in a *single-member* district. Some courts have held that this prong is inapplicable where no single-member structure exists [6]. The law on this point is unsettled.

5.4 Fair Representation Act

The Fair Representation Act (FRA), introduced in multiple Congresses by Rep. Don Beyer and others, would require states with more than one congressional seat to use multi-member districts with ranked-choice voting (STV) [1]. The Act would supersede 2 U.S.C. § 2c and explicitly authorize districts of 3–5 members. It would also amend the VRA to clarify that minority representation in MMDs satisfies Section 2 obligations.

The computational framework developed in this paper is directly applicable to implementing the FRA: bisect generates the district boundaries, and our STV implementation provides the seat-allocation algorithm. The case studies presented here could serve as illustrative plans for congressional testimony or commission evaluation.

6 Conclusion

We have demonstrated that the bisect recursive bisection algorithm extends naturally to multi-member district generation by adjusting the district-count parameter k and adapting the per-seat balance tolerance formula. Across five configurations in three states, MMDs systematically improve minority representation relative to enacted single-member plans—California’s 4-member OLPR districts produce an estimated 22 minority-opportunity seats versus 17 in the enacted plan—while incurring a modest compactness penalty of approximately 5–12% in mean Polsby-Popper depending on district magnitude.[†]

STV vs. OLPR. The two allocation rules produce similar seat distributions. OLPR produces one additional minority-opportunity seat in the California case study, driven by the direct candidate-preference mechanism. STV has the advantage of requiring no party registration, making it more accessible for jurisdictions with strong traditions of cross-party voting.

Proportionality-compactness tradeoff. The tradeoff is real but modest: the 4-member configuration achieves Gallagher indices of 1.8–2.9 (comparable to Nordic PR systems) at a 10–12% compactness cost. The 2-member configuration achieves intermediate results in both dimensions. The choice between magnitudes is primarily a values judgment—how much compactness to sacrifice for proportionality—rather than an algorithmic question.

Legal path. The statutory barrier to MMDs (2 U.S.C. § 2c) is a legislative choice, not a constitutional requirement. The Fair Representation Act would clear this barrier and

explicitly authorize the type of MMD plan generated in this paper. The 110-year Illinois precedent with three-member Cumulative Voting districts demonstrates that MMDs are compatible with American political culture.

Future work. Several extensions are warranted. First, ensemble analysis (multiple seeds) would characterise the distribution of possible MMD configurations rather than the single-run results reported here. Second, the full 50-state analysis would establish national baselines for proportionality and minority representation under various MMD designs. Third, combined MMD and VRA edge-weighting experiments would explore whether geographic minority clustering and proportional allocation are complementary or substitutable strategies.

References

- [1] 116th Congress. Fair representation act, h.r. 4000. U.S. House of Representatives, 2019.
- [2] Micah Altman and Michael P McDonald. Bard: Better automated redistricting. *Journal of Statistical Software*, 42(4):1–28, 2011.
- [3] Michel L Balinski and H Peyton Young. *Fair representation: Meeting the ideal of one man, one vote*. Yale University Press, 1982.
- [4] Richard L Cole. *Cumulative voting in Illinois, 1870–1980*. University of Illinois Press, 2001.
- [5] Giovanni Della-Libera. Edge-weighted graph partitioning for congressional redistricting. *Working paper*, 2026.
- [6] Richard L Engstrom. Minority representation under multi-member proportional systems: A voting rights act analysis, 2010. Working paper, National Conference of State Legislatures.
- [7] Michael Gallagher. Proportionality, disproportionality and electoral systems. *Electoral Studies*, 10(1):33–51, 1991.
- [8] Michael Gallagher. The results analysed. In Michael Gallagher and Michael Marsh, editors, *How Ireland Voted 2011*, pages 126–168. Palgrave Macmillan, 2011.
- [9] Michael Gallagher. The election and the party system. *Electoral Studies*, 38:98–109, 2015.
- [10] Lani Guinier. *The tyranny of the majority: Fundamental fairness in representative democracy*. Free Press, New York, 1994.
- [11] Lauri Karvonen. Preferential voting: Incidence and effects. *International Political Science Review*, 25(2):203–226, 2004.

- [12] Anthony J McGann, Charles Anthony Smith, Michael Latner, and Alex Keena. *Gerrymandering in America: The House of Representatives, the Supreme Court, and the Future of Popular Sovereignty*. Cambridge University Press, 2016.
- [13] MIT Election Data + Science Lab. 2020 u.s. presidential election returns. <https://dataverse.harvard.edu/dataset.xhtml?persistentId=doi:10.7910/DVN/VOQCHQ>, 2021.
- [14] Mark E Rush. *Redistricting and representation: Why competitive elections are bad for America*. University Press of America, 2000.
- [15] Nicholas O Stephanopoulos and Eric M McGhee. Partisan gerrymandering and the efficiency gap. *University of Chicago Law Review*, 82:831–900, 2015.
- [16] Rein Taagepera. *Predicting party sizes: The logic of simple electoral systems*. Oxford University Press, 2007.
- [17] Abigail M Thernstrom. *Whose votes count? Affirmative action and minority voting rights*. Harvard University Press, 1987.
- [18] Thornburg. *Thornburg v. gingles*, 478 u.s. 30, 1986.
- [19] Wesberry. *Wesberry v. sanders*, 376 u.s. 1, 1964.